Rational-slope rigidity with identity holonomy through finite-amplitude resonant cells

Technical summary Robert Jenkinson Alvarez

Core idea. A foliation may return to itself without returning each rational leaf to itself. That distinction separates an explicit all-rotation counterexample from rigidity. Once rational leaves close in their primitive classes, their actions become phase-flat rational cell energies. The paper classifies this finite-amplitude flatness exactly: it is equivalent to multiplicative separability of a periodic quadratic coefficient. Applied to the original equation, the classification forces $V_u \equiv 0$.

Two rigidity theorems. Theorems 1.2 and 1.4

For $u'' = -V_u(x,u)$ with $V \in C^\infty(\mathbb{T}^2)$, a class-A minimizing foliation at every primitive rational rotation $-m/\ell$, with

$$U_{m,\ell}(x+\ell,t) = U_{m,\ell}(x,t) - m, \qquad \gcd(m,\ell)=1, \quad \ell>0, \qquad \Longrightarrow \qquad V_u \equiv 0.$$

No irrational foliation, regularity in rotation, or transverse differentiability is assumed.
Independently, for *every* smooth positive $a(x,c)$ on $\mathbb{T}^2$, define the gauge-fixed rational energy

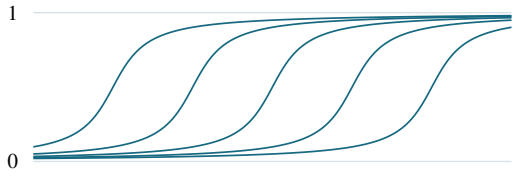
$$E_{m,\ell}(Y) = \inf_{\substack{c(x+\ell)=c(x)-m \\ \Pi_{m,\ell}(c)=Y}} \frac{1}{2\ell}\int_0^\ell a(x,c)c'^2\,\mathrm{d}x, \qquad \Pi_{m,\ell}(c) = \frac{1}{\ell}\int_0^\ell (mx+\ell c(x))\,\mathrm{d}x \pmod{\ell}.$$

With A, B smooth, positive and periodic, the exact classification is

$$\boxed{\text{Every primitive } E_{m,\ell} \text{ is independent of } Y \iff a(x,c) = A(x)B(c) \iff (\log a)_{xc} = 0.}$$

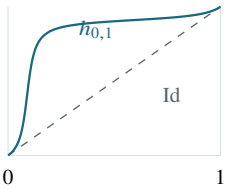
The zero-rotation obstruction

Bounded heteroclinics between integer leaves



$u_s(x) = \frac{1}{2} + \pi^{-1}\arctan(\pi(x-s))$

Primitive return map



$h_{0,1} = \Phi_1 \neq \mathrm{Id}$

1. Why all rotations do not suffice. The action is $\mathcal{A}_V[u] = \int(\frac{1}{2}u'^2 - V(x,u))\,\mathrm{d}x$; class-A means minimizing on *every* compact interval against all fixed-endpoint Sobolev competitors. For

$$V_*(u) = -\tfrac{1}{2}\sin^4(\pi u), \qquad g_E(u) = \sqrt{\sin^4(\pi u) + 2E},$$

every branch $u' = \sigma g_E(u)$, $\sigma = \pm 1$, is calibrated:

$$\mathcal{A}_{V_*}[w] - \mathcal{A}_{V_*}[u] = \tfrac{1}{2}\int (w' - \sigma g_E(w))^2\,\mathrm{d}x \geq 0.$$

The period $T(E) = \int_0^1 g_E^{-1}$ decreases from infinity to zero, so positive energies realize all nonzero rotations $\sigma/T(E)$. At zero energy, the flow of $u' = \sin^2(\pi u)$ gives the bounded leaves above. Its nonidentity return is intrinsic: the only periodic class-A minimizers are integer constants, so no alternative closed zero-rotation foliation exists.

2. What identity holonomy supplies. For rotation $-m/\ell$, ordinary invariance allows

$$U(x+\ell,t) = U(x, h_{m,\ell}(t)) - m.$$

Relabelling only conjugates $h_{m,\ell}$. Requiring $h_{m,\ell} = \mathrm{Id}$ closes each leaf, not merely the family. Ordered finite differences are then periodic positive solutions of a linear equation. Jacobi-field uniqueness and a Dirichlet implicit-function argument recover smooth transverse dependence and $U_t > 0$ from a merely continuous foliation.

3. The zero foliation becomes a coordinate. Write the closed zero foliation as $u = U_0(x,c)$, with $U_{0,c} > 0$. Its leaf field b_0 defines a closed Hilbert form

$$\omega_0 = b_0\,\mathrm{d}u - (\tfrac{1}{2}b_0^2 + V)\,\mathrm{d}x.$$

For an arbitrary graph $u(x) = U_0(x, c(x))$, square completion gives

$$(\tfrac{1}{2}u'^2 - V)\,\mathrm{d}x = \omega_0|_{\mathrm{graph}\,u} + \tfrac{1}{2}a(x,c)c'^2\,\mathrm{d}x, \qquad a = U_{0,c}^2.$$

The closed-form term cancels in fixed-endpoint and same-homology comparisons. Identity zero holonomy makes a periodic on the *product* torus. Without it, one obtains the different law

$$a(x+1,c) = a(x, h(c))h'(c)^2,$$

which is a quadratic density on a mapping torus.

4. Why the phase gauge is indispensable. The resonant variable has the unique decomposition

$$mx + \ell c(x) = Y + \varphi(x), \qquad \int_0^\ell \varphi(x)\,\mathrm{d}x = 0.$$

Allowing a free constant in φ would absorb Y and make the cell value flat for *every* a. The fixed mean removes that degeneracy. The other closed rational foliations become full-winding minimizers in the zero coordinate, and their phase means cover every Y. Consequently every genuine gauge-fixed cell is flat.

From rational cell flatness to a global rigidity theorem

5. Flat cells produce calibrated dynamics. A constrained cell minimizer initially solves

$$-(ac')' + \tfrac{1}{2}a_c c'^2 = \lambda.$$

If $E_{m,\ell}(Y)$ is constant, each constrained minimum is already a minimum over the *entire* winding class; thus $\lambda = 0$. Min–max splicing orders these minimizers, and the fixed phase makes them unique. Their family fills the configuration plane.

For $H(x, c, p) = p^2/(2a)$, the rational momentum graph $p = P_\rho(x, c)$ carries the closed form

$$\lambda_\rho = P_\rho \, \mathrm{d}c - H(x, c, P_\rho) \, \mathrm{d}x.$$

The Fenchel gap is exactly

$$L_a - \lambda_\rho|_{\text{graph}\, c} = \frac{(ac' - P_\rho)^2}{2a}.$$

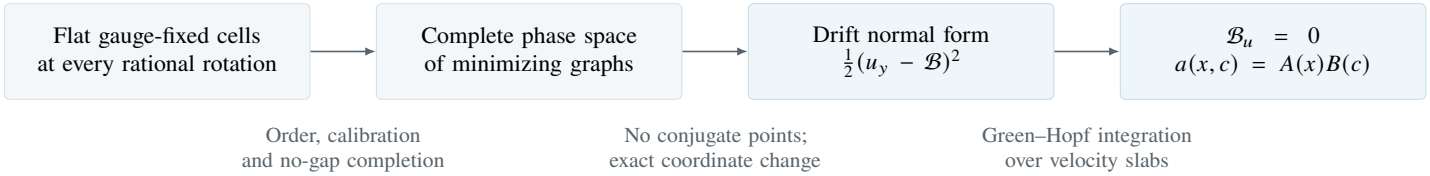
Hence every finite segment of every rational graph orbit is class-A minimizing, not just minimizing under a cell constraint.

6. Completing phase space without slope regularity. Rational graphs are strictly ordered and have uniform Lipschitz bounds on bounded rotation sets, independent of the rational denominator. Their one-sided limits remain invariant and minimizing. A calibrated quadratic comparison eliminates rational gaps. At irrational rotation, circle translation numbers eliminate disjoint gaps; for touching boundaries, area preservation rules out infinitely many disjoint images of a positive-area gap component.

An explicit momentum bound gives properness as $\rho \to \pm\infty$. Thus each fibre map $\rho \mapsto P_\rho(x, c)$ is an increasing homeomorphism and

$$\mathbb{T}^2 \times \mathbb{R}_p = \bigsqcup_{\rho \in \mathbb{R}} \Gamma_\rho.$$

Every Hamiltonian trajectory is therefore minimizing. Nonnegative index forms on all intervals, together with strict Dirichlet domain monotonicity, exclude conjugate points.



7. An exact drift normal form. For the arbitrary positive coefficient a, set

$$I(x) = \int_0^1 \sqrt{a(x, s)} \, \mathrm{d}s, \qquad S = \int_0^1 I(x)^{-2} \, \mathrm{d}x,$$

$$F(x, c) = I(x)^{-1} \int_0^c \sqrt{a(x, s)} \, \mathrm{d}s, \qquad Y_x = I(x)^{-2}/S.$$

Under $y = Y(x)$ and $u = F(x, c)$, write the drift as $\mathcal{B}(y, u) = SI(x)^2 F_x(x, c)$. Then

$$\int \tfrac{1}{2} a(x, c) c_x^2 \, \mathrm{d}x = S^{-1} \int \tfrac{1}{2}(u_y - \mathcal{B}(y, u))^2 \, \mathrm{d}y.$$

The transformation preserves winding, fixed-endpoint competitors, minimality and conjugate points. Its Euler and Jacobi equations have the form

$$u_{yy} = Q_{\mathcal{B}}, \quad Q_{\mathcal{B}} = \mathcal{B}_y + \mathcal{B}\mathcal{B}_u, \qquad J'' = KJ, \quad K = (Q_{\mathcal{B}})_u.$$

The crucial simplification is that K is independent of velocity. This makes integration over the noncompact velocity direction possible without assuming decay there.

8. The separable direction is explicit. If $a = A(x)B(c)$, the coordinate $z = \int_0^c \sqrt{B(s)} \, \mathrm{d}s$ turns the cell into a weighted Dirichlet problem. With $\beta = \int_0^1 \sqrt{B}$,

$$E_{m,\ell}(Y) = \frac{m^2 \beta^2}{2\ell \int_0^\ell A(x)^{-1} \, \mathrm{d}x},$$

independent of Y. Conversely, every nonseparable coefficient is detected by at least one primitive rational cell with unequal energies at two phases.

9. The Green–Hopf mechanism. On $\mathbb{T}^2 \times \mathbb{R}_v$, the drift flow preserves $\mathrm{d}y \, \mathrm{d}u \, \mathrm{d}v$. No conjugate points yields a bounded measurable Green field r satisfying

$$Dr + r^2 = K, \qquad D = \partial_y + v\partial_u + Q_{\mathcal{B}}\partial_v.$$

For the velocity-slab mass

$$\mathcal{H}(R) = \int_{\mathbb{T}^2 \times [-R, R]} r^2 \, \mathrm{d}y \, \mathrm{d}u \, \mathrm{d}v,$$

periodicity removes the integral of $K = (Q_{\mathcal{B}})_u$, while the two velocity faces give, for almost every R,

$$\mathcal{H}(R)^2 \le 2\|Q_{\mathcal{B}}\|^2_{L^2(\mathbb{T}^2)} \mathcal{H}'(R).$$

Any positive value would force blow-up at finite R. Hence $r = 0$ and $K = 0$. Along the complete drift characteristics, $w = \mathcal{B}_u$ then solves $w' = -w^2$; bounded periodicity forces $w = 0$. Since

$$\mathcal{B}_u = SI(x)^2 \frac{F_{xc}}{F_c},$$

one obtains $F_{xc} = 0$ and $a = I(x)^2 G(c)^2$.

10. Returning to the original potential. For $a = U_{0,c}^2$, separability and the vertical deck law imply $U_0(x, c) = \psi(x) + H(c)$. The zero-leaf equation becomes

$$\psi''(x) = -V_u(x, \psi(x) + H(c)).$$

Since H is onto, V_u is independent of u. Its vertical mean vanishes by periodicity of V, so $V_u \equiv 0$.

Scope. The rational-cell classification is global and finite-amplitude, with no smallness hypothesis. The original rigidity theorem needs only independently existing rational minimizing foliations with identity primitive holonomy; their transverse smoothness and the continuous phase-space completion are conclusions. Ordinary all-rotation foliations alone are insufficient. Source: Theorems 1.1, 1.2, 1.4, 8.7, 10.1 and 11.6; §13.